

QR Code Generator & Scanner using Python - Synopsis

Project Title: QR Code Generator & Scanner using Python

Objective:

The objective of this project is to create a simple Python application that can generate QR codes from input text or URLs and also scan QR codes to retrieve the encoded information.

Tools/Technologies Used:

- Python Programming Language
- qrcode library
- OpenCV (for scanning QR codes)
- NumPy
- IDE: Visual Studio Code / PyCharm

Project Description:

This project allows users to input text or URLs and generate a QR code image instantly. The generated QR code can be saved as an image file. Additionally, the scanner module uses a webcam or image input to decode QR codes and display the stored information.

Working:

1. User provides input (text/URL).
2. Python script generates a QR code image using the qrcode library.
3. For scanning, OpenCV captures image from webcam or file.
4. The QR code is detected and decoded to retrieve data.

Applications:

- Contact sharing
- Website linking
- Product tracking
- Attendance systems

Conclusion:

This project demonstrates the practical use of Python libraries in real-world applications like data encoding and decoding using QR codes.